Introduction to OOP 2023-2024 Project

Mert Kerem Dalkılınç – Yiğit Çakar | 19.05.2024
05220000277 - 05220000329

CONTENTS

Explanation of the Code……2

Class and Methods Used in the Code……2

Time Spent on Design, Implementation, and Testing……3

Conclusion……3

INTRODUCTION TO PROJECT

This project aims to create a program that manages customer data, tracks product ratings, and analyses user preferences. We use Java to work with data structures like linked lists and arrays, and we handle files to read and write data. In this section, we'll explain how the main parts of the project work and how the code accomplishes its tasks.

Class and Methods Used in the Code

1. **Main Class:**
   - **public static void main(String[] args):** It's the main entry point of the program. Initiates the menu operations.
   - **private static void readFromFile(String fileName) throws IOException:** Reads data from a file and creates data sets.
   - **private static void addNewCustomer(Scanner scanner):** Enables adding a new customer.
   - **private static void printAverageRatings():** Prints average ratings for all products.
   - **private static void printAverageRatingsForTurkey():** Prints average ratings for all products for customers from Turkey.
   - **private static void printAverageRatingsForNonTurkey():** Prints average ratings for all products for customers from countries other than Turkey.
   - **private static void printAverageRatingsForDoctors():** Prints average ratings for all products for customers whose occupation is 'Doctor'.
   - **private static void printRatingsMatrix():** Prints the two-dimensional rating matrix.
2. **CustomerData Class:**
   - **public CustomerData(String name, String surname, String country, String city, String occupation):** Constructor used to hold customer data.
   - **public String toString():** Returns customer data as a string.
3. **LinkedList Class:**
   - **public void addNode(int customerNumber, CustomerData customerData):** Adds a new customer node to the linked list.
   - **public void printList():** Prints the linked list.

- **public Node getHead():** Returns the head of the linked list.

4. **<u>Node Class:</u>**
   - **public Node(int customerNumber, CustomerData customerData):** Constructor that creates a node in the linked list.
   - **Getter and setter methods:** getCustomerNumber(), getCustomerData(), getNext(), setCustomerNumber(), setCustomerData(), setNext().

Time Spent on Design, Implementation, and Testing

Stage	Time Spent (Hour)
Design	6
Implementation	16
Testing	7

Conclusion

In conclusion, this Java program effectively manages customer data and product ratings through a user-friendly interface. It provides essential functionalities such as reading data from files, adding new customers, and calculating average ratings for products. The program's modular structure enhances code readability and maintainability, making it suitable for future enhancements or modifications. Overall, the project demonstrates proficiency in object-oriented programming concepts and Java development skills.